

①

$$dB = 10 \log \left(\frac{I}{10^{-12}} \right)$$

Orquesta $\rightarrow 80dB = 10 \log \left(\frac{I_{org}}{10^{-12}} \right)$

$$10^8 \cdot 10^{-12} = I_{org}$$

$$\underline{10^{-4} = I_{org}}$$

$$I_{vidin} = \frac{10^{-4}}{250}$$

Luego, $dB_{vidin} = 10 \log \left(\frac{10^{-4}}{250 \cdot 10^{-12}} \right)$

$$\boxed{dB_{vidin} = 56 \text{ dB}}$$

②

$$f = 450 \text{ Hz}$$

$$v_s = 340 \text{ m/s}$$

$$v_f = 30 \text{ m/s}$$

$$v_o = 0 \text{ m/s}$$

Cuando se acerca:

$$f' = f \cdot \left(\frac{v_s - v_o}{v_s - v_f} \right)$$

$$f' = 450 \left(\frac{340 - 0}{340 - 30} \right)$$

$$\boxed{f' = 493,5 \text{ Hz}}$$

Cuando se aleja:

$$f' = f \cdot \left(\frac{v_s - v_o}{v_s + v_f} \right)$$

$$f' = 450 \left(\frac{340 - 0}{340 + 30} \right)$$

$$\boxed{f' = 413,5 \text{ Hz}}$$

(2)

(3) Como $f_n = \frac{v}{2L}$

Si L disminuye, f aumenta y por eso suena más agudo.

(4) $L = 20 \text{ m}$

$f = 80 \text{ Hz}$

$n = 3$

$\lambda_n = \frac{2L}{n} = 13,3 \text{ m}$

$v = \lambda \cdot f = 13,3 \cdot 80 = 1066,6 \text{ m/s.}$ $\rightarrow$ la velocidad de propagación.

$f_2 = \frac{v}{\lambda_1} = \frac{v}{2L}$

$f_2 = 26,6 \text{ Hz}$

$f_2 = \frac{v}{\lambda_2} = \frac{v}{L}$

$f_2 = 53,3 \text{ Hz}$

(8) $L = 80 \text{ cm} \rightarrow 0,8 \text{ m}$
 $n = 3$

Abierto

$f_n = n \frac{v}{2L} = \frac{3 (340 \text{ m/s})}{2 (0,8 \text{ m})} = 637,5 \text{ Hz}$

$\lambda_n = \frac{v}{f_n}$

$\lambda_3 = 0,53 \text{ m}$

Cerrado

$f_n = (2n-1) \frac{v}{4L} = \frac{5 (340)}{4 (0,8)} = 531,25 \text{ Hz}$

$\lambda_n = \frac{v}{f_n}$

$\lambda_3 = 0,64 \text{ m}$